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Special Topic: Thermoelectric Materials and Devices

Revealing unipolar thermoelectric performance in bipolar polymer

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Abstract: Conjugated polymers are attracting increased attention as thermoelectric (TE) materials for energy harvesting applications in low-temperature regimes. However, in many doped ambipolar polymers, the simultaneous transport of both holes and electrons under temperature gradients leads to an offset in thermopower (S), which suppresses TE performance and complicates intrinsic understanding of bipolar TE conversion. Herein, we quantitatively investigate the p-n polarity transition in FeCl₃-doped bipolar PDPP4T films by measuring the magneto-thermoelectric Nernst effect, combined with Hall and Seebeck effect analyses. Notably, behind the $S = 0$ point, we observe a significant thermopower offset originating from the balancing contributions of electrons and holes. This countervailing thermopower value is extracted to reach 400 $\mu\text{V K}^{-1}$, which could ideally produce an estimated maximum unipolar ZT of 0.24 at 175 K, due to rising polaron states and reduced carrier concentration. Our findings reveal the extraordinary hidden unipolar TE performance achievable in doped bipolar polymer towards ultra-low-temperatures thermoelectric.

Keywords: organic thermoelectric materials, transverse thermoelectrics, Nernst effect, bipolar polymer, ultra-low-temperature thermoelectrics

INTRODUCTION

Polymeric semiconductor-based thermoelectric (TE) materials are gaining expanding attention for their potential to enable flexible devices capable of directly converting heat into electricity, making them ideal for powering wearable electronic devices and the Internet of Things. To date, most reported polymeric TE materials are regarded to be unipolar candidates. In fact, bipolar charge transport is commonly observed in conjugated polymers, which can be manipulated via tuning the interfacial defects [1,2], field-effect [3–5], temperature [6,7], and doping level to create functional devices [8–11], such as organic transistors and organic TE devices, etc. From a TE perspective, the polarity of polymers is typically determined by the sign of the thermopower (S), except at the notable p-n switching point where S equals zero [6–11]. This Seebeck

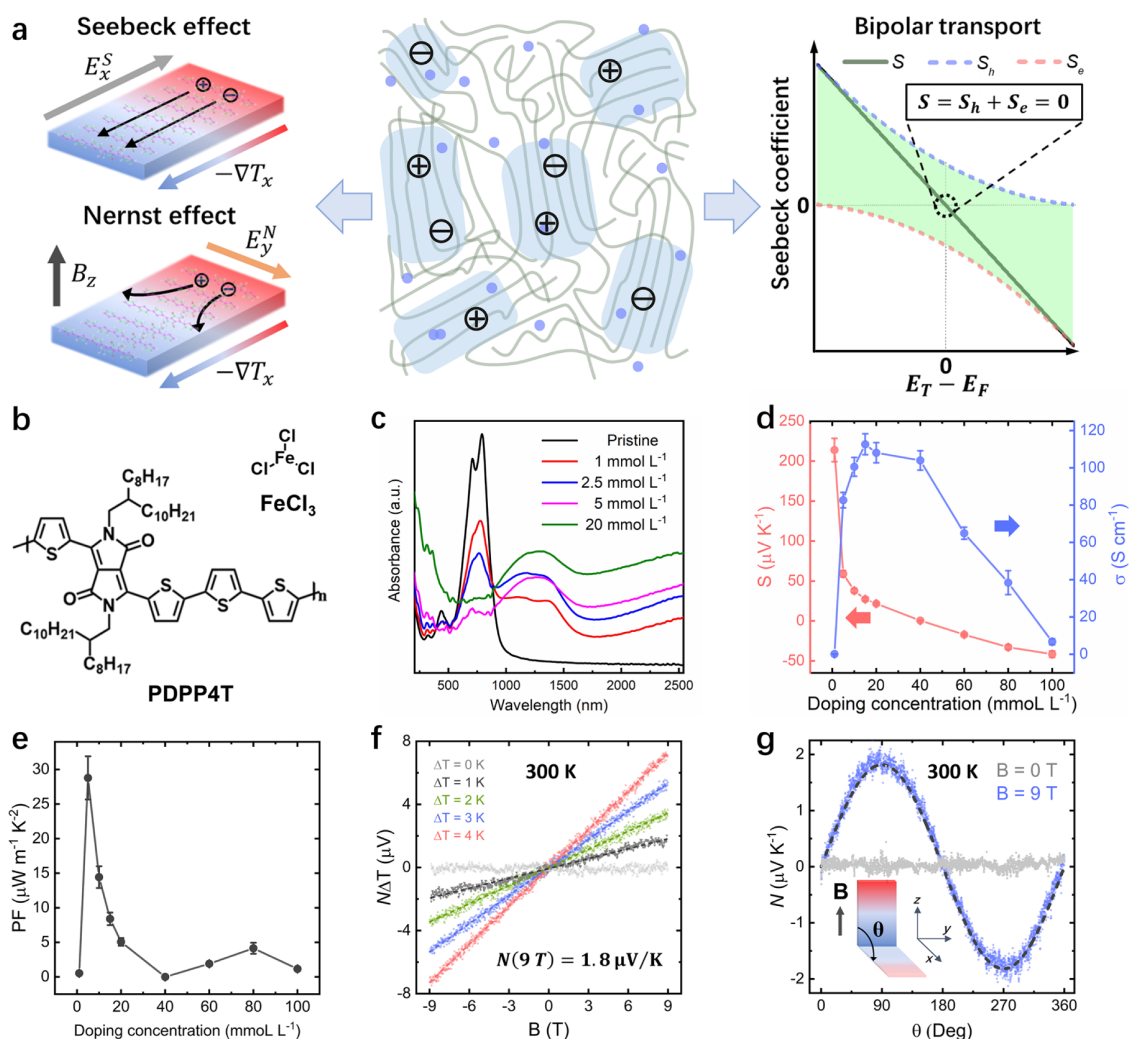


Figure 1 Bipolar transport and Nernst effect measurements in polymer at 300 K. (a) Schematic illustration of general bipolar charge transport in doped polymer (middle panel), in which the greyish-green lines represent polymer chains, the gray-blue shaded areas represent crystalline regions, and the bluish-violet circles represent the dopant counterions; schematic of bipolar Seebeck effect with longitudinal temperature gradient and the transverse Nernst effect with an additional perpendicular magnetic field in polymer devices (left panel); schematic plot of thermopower S (dark green line) polarity transition versus the difference between transport level and Fermi level $E_T - E_F$ in polymers (right panel), where the $S = 0$ transition point is indicated by the dashed circle, while the dashed blue and red curves and electron partial thermopower, respectively. (b) Molecular structure of polymer PDPP4T and dopant FeCl_3 . (c) Ultraviolet-visible-near-infrared (UV-vis-NIR) absorption spectra, (d) thermopower and electrical conductivity, and (e) power factor of PDPP4T films doped with different FeCl_3 concentrations. (f) Transverse Nernst effect voltage signal versus magnetic field under different temperature gradient in 20 mmol L⁻¹ FeCl_3 -doped PDPP4T device. The colored lines correspond to linear fits. (g) PDPP4T (20 mmol L⁻¹) Nernst coefficients versus the angle θ between the temperature gradient ($\Delta T = 4 \text{ K}$) and the magnetic field direction when $B = 0$ and $+9 \text{ T}$. The dotted black curve plots the sine function. The device was rotated in the xz plane while the magnetic field was fixed in the z direction.

polarity transition can be schematically represented by variations between the transport level E_T and the Fermi level E_F (Figure 1a, Supplementary Discussion S1) [12]. In non-superconducting systems, $|S|$ is larger than zero for either electrons or holes when $T > 0 \text{ K}$ due to the transport entropy of thermally excited charge carriers [13–15]. Accordingly, the observed zero S should originate from the offset of positive hole and negative electron contributions. It therefore arises a key open question: what is the quantitative “true” unipolar TE performance behind zero S in the bipolar regime of doped polymers?

To answer this long-sought challenging issue, it is essential to separate the electron and hole contributions in bipolar TE conversion. Notably, a similar conundrum in inorganic systems has been ingeniously overcome by combining the magneto-thermoelectric Nernst, Seebeck, Hall, and magnetoresistance (MR) effects with appropriate approximations [16–18]. Despite these inspiring efforts, this methodology has not translated to polymeric semiconductors, as organic conjugated materials are aggregated by van der Waals interactions and exhibit distinct charge transport mechanisms compared to inorganic counterparts. More importantly, the transverse Nernst effect [16–21], which is essential for distinguishing electron and hole contributions in TE transport, has rarely been explored in organic systems due to the weak signal-to-noise ratio resulting from their inherent low mobility and conductivity [22]. These combined challenges prevent the isolation of unipolar thermopower, thereby hindering a deeper understanding of TE conversion in bipolar-doped polymers.

Herein, we quantitatively reveal the thermoelectric properties of electrons and holes in bipolar FeCl₃-doped PDPP4T films across different temperatures by conducting the Nernst effect measurements, along with corresponding Seebeck and Hall effect analyses. Notably, at low temperatures around the $S = 0$ point, the hole and electron power factors (PF) are significantly enhanced, reaching 145.9 and 83.1 $\mu\text{W m}^{-1} \text{K}^{-2}$, respectively, due to increased energetic disorder, reduced carrier density, and hole-electron Coulomb interactions. Combined with a reduction in thermal conductivity at low temperatures, these effects ideally could yield remarkable maximum ZT values of 0.24 for holes and 0.13 for electrons at 175 K, setting a new benchmark for ultra-low-temperature thermoelectrics. This is particularly notable as current commercial TE materials typically operate at or above room temperature, while there is a growing need for ultra-low-temperature TE applications in environments ranging from -150 to 0°C .

RESULTS AND DISCUSSION

Different from the counteractive contributions of holes and electrons in the Seebeck effect, a magnetic field perpendicular to the temperature gradient in the Nernst effect deflects holes and electrons into opposite directions due to the Lorentz force, leading to an accumulated transverse voltage (Figure 1a). Therefore, combining the Seebeck and Nernst effects could theoretically separate the electron and hole TE contributions in the bipolar regime of doped polymers [16–18,23,24], with additional information of charge mobilities which can be acquired by the Hall effect measurements [25,26]. Specifically for polymer films, this can be achieved by setting up and solving isotropic Boltzmann transport equations in the low-mobility regime ($\mu B \ll 1$) with appropriate approximations (Supplementary Discussions S2 and S3), as listed in Eqs. (1)–(4):

$$S = \frac{\sigma_h S^h + \sigma_e S^e}{\sigma}, \quad (1)$$

$$v = \frac{\sigma_h v_h + \sigma_e v_e}{\sigma} + \frac{(\mu_h + \mu_e) \sigma_h \sigma_e (S^h - S^e)}{\sigma^2}, \quad (2)$$

$$\sigma = \sigma_h + \sigma_e, \quad (3)$$

$$R_H = \frac{(\sigma_h \mu_h - \sigma_e \mu_e)}{\sigma^2}. \quad (4)$$

where the v, σ, R_H, μ denote the Nernst coefficient, conductivity, Hall coefficient and mobility, respectively.

The e and h subscripts indicate the corresponding electron and hole components. For a more physical and direct relation, the holes and electrons partial thermopower S_h and S_e illustrated in Figure 1a are defined as $S_{h/e} = S^{h/e} \sigma_{h/e} / \sigma$, so that $S = S_h + S_e$, where $S^{h/e}$ and $\sigma_{h/e}$ are corresponding hole and electron Seebeck coefficients and electrical conductivities, respectively.

As discussed above, to establish a reliable approach for quantitatively exploring unipolar TE properties in bipolar regime of doped polymers, the first crucial step is to observe and accurately measure the Nernst effect in doped polymers. For this study, we selected FeCl₃-doped PDPP4T as a well-characterized bipolar system (Figure 1b). While challenging p-type and n-type Hall effect measurements have been successfully conducted on FeCl₃-doped PDPP4T [8], its relatively low conductivity poses limitations for the more demanding Nernst effect measurements, which are analogous to thermoelectric Hall effect measurements but involve much lower longitudinal currents driven by the Seebeck effect. To address this issue, we employed a sequential doping method to enhance PDPP4T's electrical conductivity, thereby reducing channel resistance and noise levels in the Nernst effect measurements. The high sequential doping and charge transfer efficiency is demonstrated by the ultraviolet-visible-near infrared (UV-vis-NIR) absorption spectra (Figure 1c) and the ultraviolet photoelectron spectroscopy/low energy inverse photoemission spectroscopy (UPS/LEIPS) characterizations (Figure S1) upon different FeCl₃ doping concentrations [27–29]. Moreover, the grazing incidence wide angle X-ray scattering (GIWAXS) characterizations reveal similarly high crystallinity for the sequential FeCl₃-doped PDPP4T films as compared to the pristine case (Figure S2). With increasing doping levels, consistent PDPP4T Seebeck coefficients sign switch phenomenon from positive to negative with the previous study is observed [8], with much higher electrical conductivity over 100 S cm⁻¹ for FeCl₃ doping concentration between 10 and 20 mmol L⁻¹ (Figure 1d). The maximum p-type and n-type PFs are 28.7 and 1.9 μW m⁻¹ K⁻² occurred at FeCl₃ doping concentration of 5 and 80 mmol L⁻¹ (Figure 1e), respectively. It should be noted that the PDPP4T p-n transition point was identified at a FeCl₃ doping concentration of 40 mmol L⁻¹, where a nearly zero S and PF were observed.

With enhanced film conductivity, noise in the PDPP4T Nernst measurements was minimized through optimization in device geometry, patterning, encapsulation, and the custom-made setup (see Figure S3 and MATERIALS AND METHODS section). The temperature calibration and thermoelectric stability tests of the Nernst device were systematically performed as well (Figures S3 and S4). Figure 1f shows the linear magnetic field (−9 to +9 T) and temperature gradient (1 to 4 K) dependent transverse Nernst voltages in 20 mmol L⁻¹ FeCl₃ doped PDPP4T device measured at 300 K. In contrast to the negligible signal without a temperature gradient, the Nernst coefficients $N = E_y / \nabla T_x$ and $\nu = N/B$ were measured as 1.8 μV K⁻¹ and 0.2 μV K⁻¹ T⁻¹, respectively. Of particular note, the effects of MR, dopants, and doping method on the Nernst signal were all carefully considered and excluded (Figures S5 and S6). The most conclusive evidence for the Nernst effect observation is its magnetic-field angular dependence. As the device was rotated in the xz plane, a perfect sine relation between N and angle θ was revealed (Figure 1g). The result decisively demonstrates the observed transverse magneto-thermoelectric voltage signal originated from Lorentz force ($\nu \times B$) exerted on the charge carriers in doped PDPP4T, thus verifying the nature of the Nernst effect. Furthermore, no signal was detected between N and θ without a magnetic field under the same temperature gradient, which provides additional confirmation for the measured Nernst effect.

The successful observation of the Nernst effect enables a quantitative investigation of bipolar TE conversion in doped PDPP4T. To avoid potential systematic errors introduced by device variation in manip-

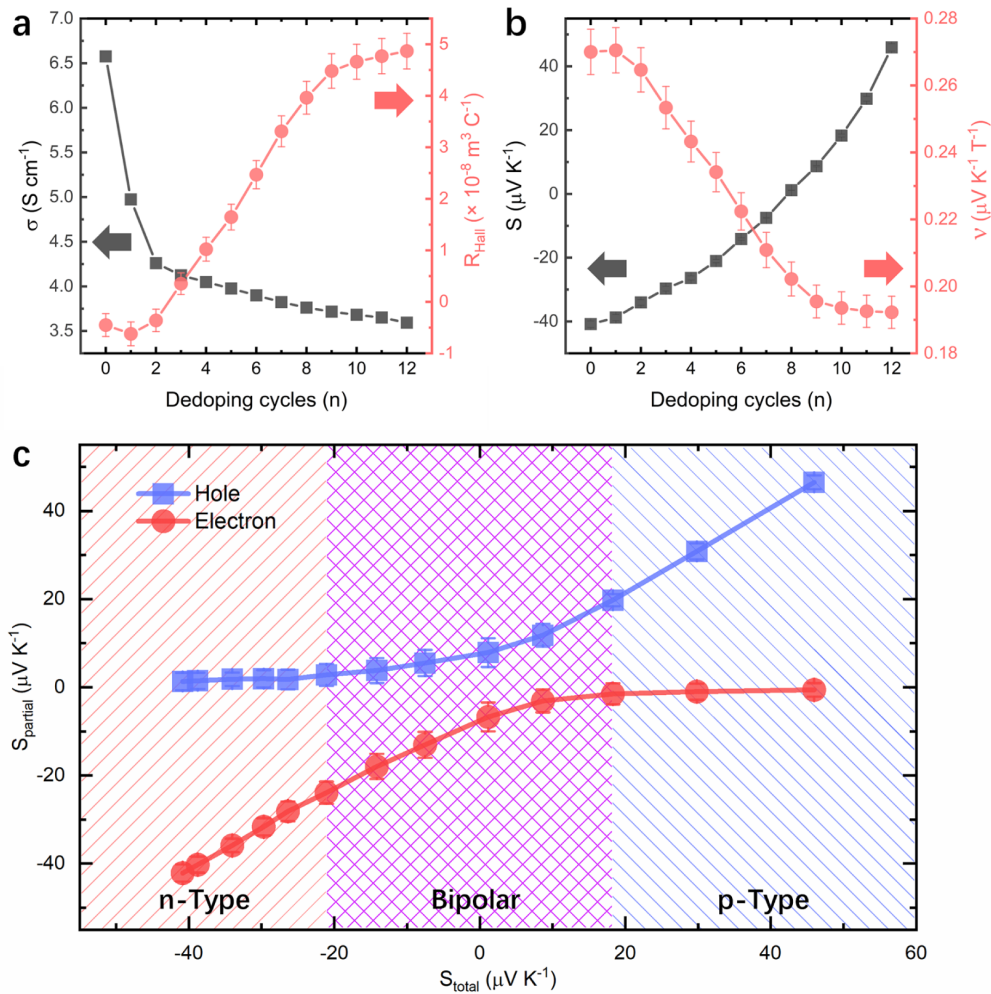


Figure 2 Evolution of the Seebeck p-n transition at 300 K. (a) Variations of the conductivity and Hall coefficients, and corresponding. (b) Seebeck and Nernst coefficients during the *in-situ* dedoping process of one a highly doped (100 mmol L⁻¹ FeCl₃) PDPP4T device. The dedoping was achieved by repeating a specific annealing process under vacuum (see MATERIALS AND METHODS section), and after each annealing process the device will cool down to 300 K to perform the conductivity, Hall, Seebeck, and Nernst effect measurements. The number of dedoping cycles represents the accumulated number of annealing process. (c) Revealed PDPP4T hole and electron partial thermopower evolution versus total thermopower during the entire dedoping process from n-type to p-type.

ulating Seebeck polarity transitions, *in-situ* characterization on a single device provides a more reliable approach. For this purpose, we conducted *in-situ* dedoping studies at 300 K on a highly doped (100 mmol L⁻¹) PDPP4T device [30], allowing it to pass through the polarity transition point. The dedoping is achieved via repeating certain annealing cycles. During each cycle, the highly doped PDPP4T device is annealed at 350 K under vacuum ($<10^{-2}$ Torr) for 60 min. The Seebeck, conductivity, Hall, and Nernst measurements are performed after each annealing cycle when the PDPP4T device temperature is decreased to 300 K. As dedoping progresses, the conductivity of PDPP4T decreases (Figure 2a). Initially, the Hall coefficient R_{Hall} is negative, indicating n-type behavior, but it switches to a positive value, indicating a shift to p-type behavior. With additional dedoping cycles, the positive R_{Hall} continues to increase, reflecting both a reduced doping level and a decrease in Hall carrier concentration n_{Hall} ($R_{\text{Hall}} = 1/(n_{\text{Hall}}e)$) (Figure S7). For the Seebeck and Nernst effects, dedoping induces a gradual change in thermopower, shifting from n-type

($-40.8 \mu\text{V K}^{-1}$) to p-type ($46.0 \mu\text{V K}^{-1}$), while the ν decreases from n-type ($0.27 \mu\text{V K}^{-1} \text{T}^{-1}$) to dedoped p-type ($0.17 \mu\text{V K}^{-1} \text{T}^{-1}$) (Figure 2b and Figure S8).

Combined Hall, Seebeck, and Nernst effect measurements allow for quantitative analysis of the unipolar charge properties in bipolar PDPP4T (Figures S7–S10). During the dedoping process of 100 mmol L^{-1} FeCl_3 -doped PDPP4T, although the overall carrier concentration is reduced, the hole concentration is found to be increased. Specifically, the electron Hall concentration decreases dramatically from 9.0×10^{20} to $1.3 \times 10^{19} \text{ cm}^{-3}$, while the hole Hall concentration increases from 1.4×10^{19} to $1.2 \times 10^{20} \text{ cm}^{-3}$ (Figure S11), coinciding with the transition from negative to positive Seebeck coefficients. More importantly, regarding the TE conversion, the evolution of holes and electrons partial thermopower (S_h , S_e) during the p-n transition is now fully revealed, as plotted in Figure 2c. The result shows that the entire polarity transition process can be identified into three regimes: the electron dominating, hole dominating, and bipolar transport regimes. In particular, within the bipolar regime between $S = -21.1$ – $18.3 \mu\text{V K}^{-1}$, the hole and electron Seebeck contributions are comparable ($|S_h| < 10 \times |S_e|$ or vice versa). Despite the total thermopower being close to zero as for the bipolar regime, clear contributions from both hole and electron Seebeck effects are still observed.

After revealing the entire p-n transition process at 300 K, we focused on the Seebeck p-n switching point ($S \sim 0$) and performed *in-situ* temperature-dependent (from 300 to 50 K) measurements in a 40 mmol L^{-1} FeCl_3 -doped PDPP4T device. During the cooling process, the electrical conductivity of PDPP4T decreases, following typical semiconducting and thermally activated behavior (Figure 3a). Meanwhile, the Hall coefficient R_{Hall} increases, indicating a reduction in n_{Hall} (Figure S12), likely due to reduced charge transfer efficiency between dopants and molecules or enhanced carrier localization at low temperatures. Over the entire temperature range, the Seebeck coefficients exhibit only small variations around zero, while the Nernst coefficients of PDPP4T show unusual nonmonotonic temperature-dependent behavior (Figure 3b and Figure S12). Specifically, ν increased from $0.22 \mu\text{V K}^{-1} \text{T}^{-1}$ at 300 K to $0.35 \mu\text{V K}^{-1} \text{T}^{-1}$ at 175 K, then began to decrease with further cooling. A similar temperature dependence of the Nernst effect was observed in other doped PDPP4T devices (Figure S9). From the above measurements, nearly equal Seebeck coefficients (S_h and S_e) with opposite signs were revealed over a broad temperature range (Figure 3c). More importantly, in contrast to the nearly unchanged total thermopower, both S_h and S_e display unexpected significant and nonmonotonic temperature dependence, which can be separated into two stages. In the temperature regime from 300 to 175 K, both S_h and $|S_e|$ are greatly enhanced, followed by a decrease for temperatures below 175 K. In particular, S_h and S_e reach maxima of 197.5 and $-201.3 \mu\text{V K}^{-1}$, respectively, at 175 K, with a total S of $-3.8 \mu\text{V K}^{-1}$. Given that $S_{h/e} \sim \nu/\mu$ for nearly perfect hole/electron balance in this case (Supplementary Discussion S2 and Eqs. (21) and (22) in the Supplementary information), the enhanced unipolar thermopower can be mathematically understood from the increased Nernst coefficients while decreased charge mobilities at low temperatures (Figure 3b and Figure S9c).

Regarding the potential errors and uncertainties in our analysis of unipolar thermoelectric (TE) properties via solving the Boltzmann transport equations, one key factor is the determination of charge mobilities in doped PDPP4T. Hall mobilities are often considered underestimated compared to intrinsic drift mobilities due to screening effects and the presence of localized charge carriers. However, to the best of our knowledge, there seems to be no other mobility measurement methods for doped polymers which have been demonstrated to be more accurate than the Hall effect. On the other hand, even if Hall mobilities are underestimated, they should still be regarded as the relevant “effective” mobilities for solving the Boltzmann equations that

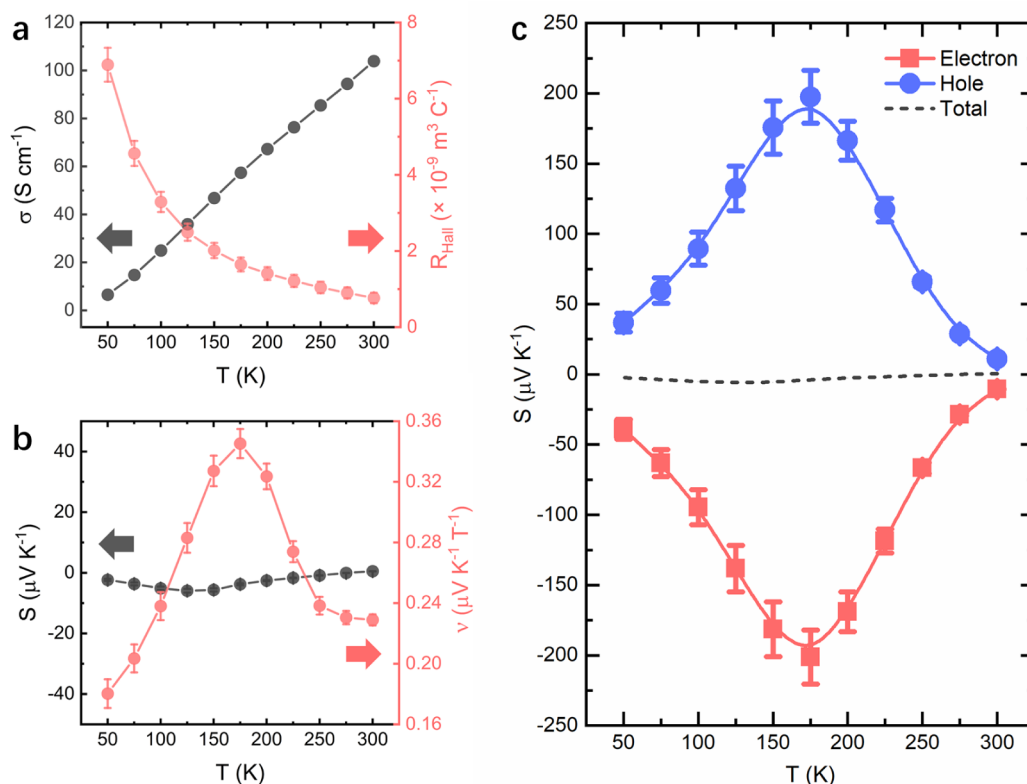


Figure 3 Temperature-dependent investigation of the Seebeck p-n switching ($S = 0$) point. *In-situ* variable temperature (300 to 50 K). (a) Conductivity and Hall effect, and (b) Seebeck and Nernst effect measurements in 40 mmol L⁻¹ FeCl₃-doped bipolar PDPP4T device. (c) Revealed nonmonotonic temperature dependence of hole and electron partial thermopower in bipolar PDPP4T. The dashed black line represents the measured bipolar thermopower in (b).

include the Nernst effect [22]. This is because both the Hall and Nernst effects are magnetic transport phenomena of the charge carriers in polymer films, fundamentally due to the same mechanism of Lorentz force. Any physical process, such as screening, which affects the Hall coefficients should also similarly impact the Nernst coefficients. Another potential source of uncertainty is the use of the Boltzmann transport model. While the model has been successfully employed to describe TE conversion in polymers [12,15,19], it is essentially a semiclassical approach designed primarily for inorganic crystalline systems and does not account for nontrivial charge scattering mechanisms present in polymers. Nevertheless, since the Nernst effect can be interpreted as a thermoelectric Hall effect, the observed decrease in mobility coupled with an increase in Nernst coefficients at low temperatures suggests an enhanced TE conversion capability and larger thermopower.

To probe the underlying origin of the enhanced unipolar thermopower at low temperatures, *in-situ* temperature-dependent UV-vis-NIR, Raman, and UPS measurements were performed on both pristine and doped PDPP4T films, in combination with theoretical calculations. Notably, the UV-vis-NIR spectra reveal a clear redshift of the intramolecular charge transfer (ICT) absorption band, accompanied by an increased peak intensity ratio (I_{A_1}/I_{A_2}) as the temperature decreases (Figure 4a and Figure S13). This indicates that the ICT between the PDPP4T donor-acceptor (D-A) units is enhanced at low temperatures [31–33], as illustrated in Figure 4b. Furthermore, a shoulder peak at ~870 nm emerges in Figure 4a, which is attributed to charge

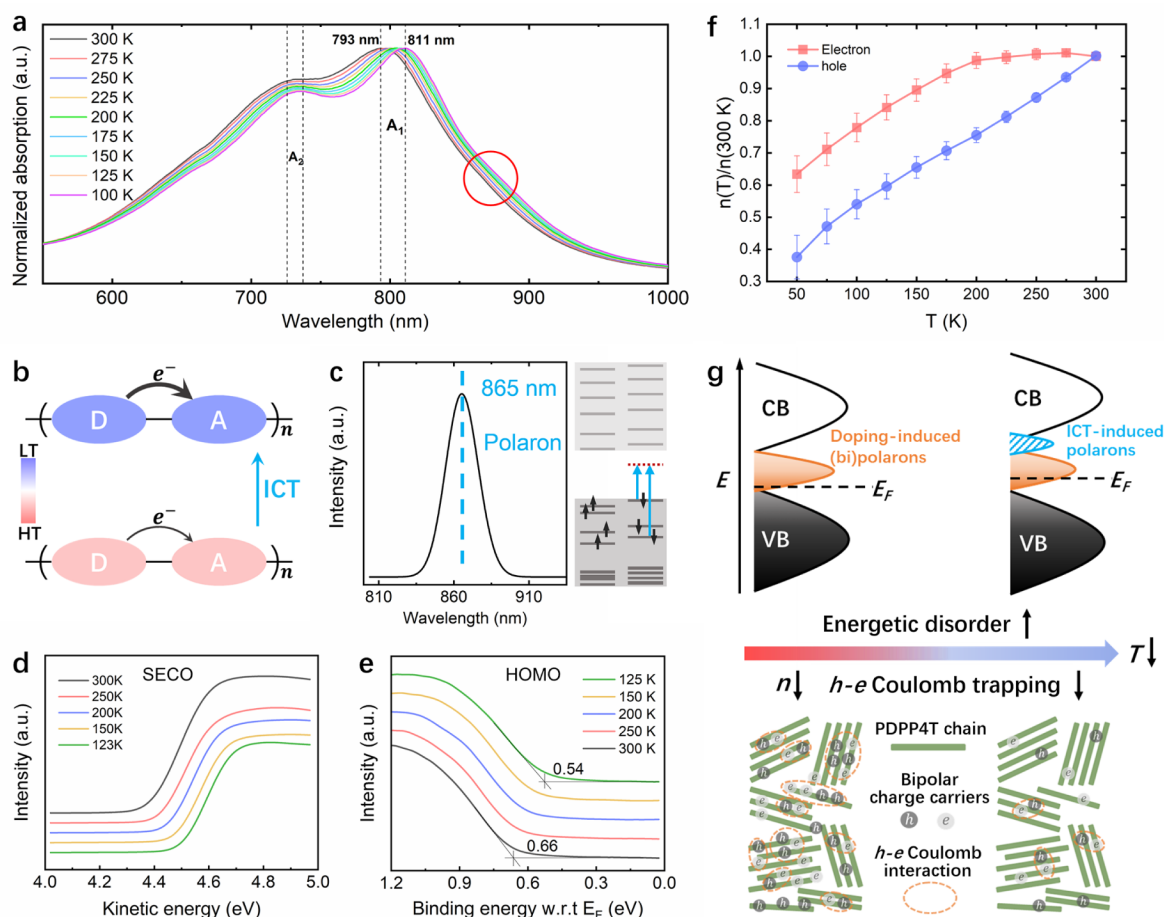


Figure 4 Mechanism for the revealed nonmonotonic thermopower temperature dependence in bipolar PDPP4T. (a) Variable temperature UV-vis-NIR absorbance of pristine PDPP4T, revealing a redshift for both A_1 and A_2 peaks, and a rising of the shoulder-peak (~ 870 nm) at a low temperature as indicated by the circle. (b) Schematic illustration of the enhanced PDPP4T donor-acceptor (D-A) intramolecular charge transfer (ICT) at low temperature, as revealed by the redshift in the absorption spectra. (c) Theoretical calculations attribute the observed rising shoulder peak in (a) at low temperatures to charge transitions to the in-gap polaronic level, as shown in the spin-split orbital energy level diagram and indicated by the blue arrows. Temperature-dependent (d) secondary electron cut-off (SECO) and (e) HOMO region of UPS characterizations for pristine PDPP4T. (f) Temperature-dependent hole and electron Hall carrier concentration in bipolar PDPP4T ($40 \text{ mmol L}^{-1} \text{ FeCl}_3$). The hole/electron concentration at low temperatures $n(T)$ is normalized to their corresponding values at 300 K. (g) Schematic illustration of the collaborative origins of enhanced thermopower at low temperature: the emergence of ICT-induced polaronic states, reduced carrier concentration, and weakened hole-electron carrier-carrier Coulomb interaction.

transitions to the in-gap polaronic level (865 nm) based on time-dependent density functional theory (TDDFT) calculations (Figure 4c and Figure S14). This is consistent with the formation of ICT-induced polarons in low-bandgap polymers [34,35]. The enhanced ICT is further supported by UPS measurements, which show an increased p-doping level at low temperatures for pristine PDPP4T, as evidenced by the Fermi level shifting toward the highest occupied molecular orbital (HOMO) level (Figure 4d and e). Additional evidence for the formation of polarons at low temperatures is provided by temperature-dependent Raman spectroscopy (Figure S15), which reveals more ordered molecular packing, likely due to reduced thermal fluctuations, as well as a polaronic signature that a shoulder peak at 1560 cm^{-1} appears as the temperature decreases [36]. On the molecular level, the enhanced ICT at low temperatures is origin from the more planarized PDPP4T backbone, as demonstrated by the molecular dynamics (MD) simulations and DFT

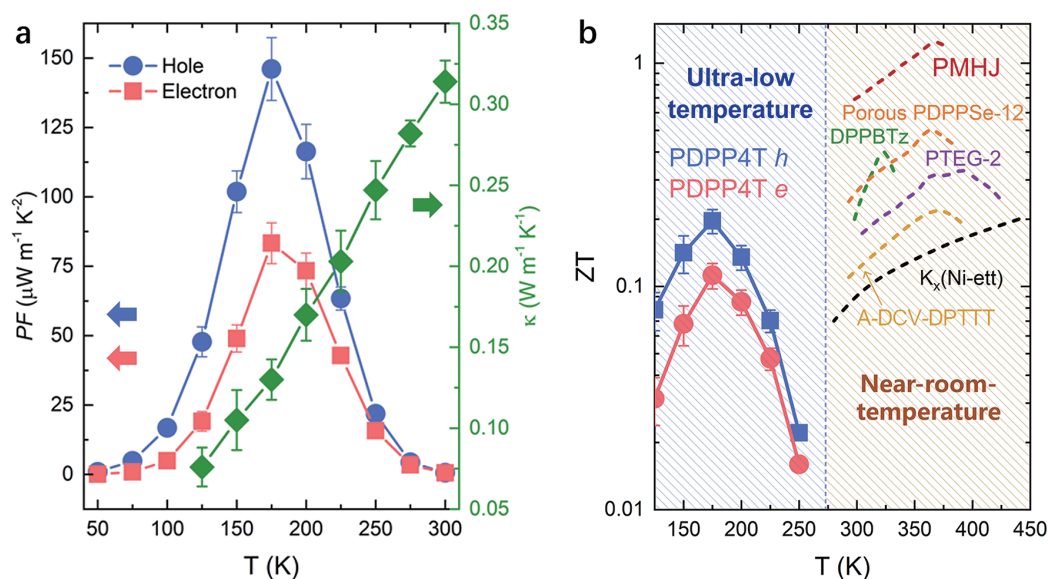


Figure 5 Unipolar TE performance in bipolar PDPP4T. (a) Hole and electron PF and measured κ_{eff} of doped (40 mmol L^{-1}) PDPP4T film at different temperatures. (b) Resulting PDPP4T unipolar ZT values in the ultra-low temperature regime (below 273 K) and comparison to reported temperature-dependent TE performance of state-of-the-art OTE materials (PMHJ [41], porous PDPPSe-12 [42], DPPBTz [43], PTEG-2 [44], A-DCV-DPTT [45], $\text{K}_x(\text{Ni-ett})$ [46]), whose largest ZT values are mostly at or near room-temperature regime.

calculations (Figure S16) [37,38], which is consistent with the Raman and other above experimental observations.

At low temperatures, the enhanced ICT-induced narrow in-gap polaronic states are expected to affect the shape of density of states (DOS) of PDPP4T around the Fermi level, leading to increased energetic disorder and thereby boosting the thermopower. This effect is particularly significant considering that FeCl_3 doping-induced (bi)polarons have been shown to govern the thermopower in doped PDPP4T [8]. Additionally, the overall doping level, as well as the extracted hole and electron concentrations, decrease at low temperatures for bipolar PDPP4T, as indicated by Hall effect and UPS analyses (Figures 3a and 4f, and Figure S17). This reduction in doping level favors an increase in thermopower, since $S \propto 1/n$. Moreover, fewer charge carriers reduce carrier-carrier Coulomb trapping between electrons and holes at low temperatures, thus freeing up more transport sites for carriers and increasing entropy [39,40]. This effect is especially pronounced in the highly doped bipolar regime, where the hole/electron components are most balanced and the carrier concentration is high. Therefore, the combined effects of increased energetic disorder due to ICT-induced polaron states, reduced carrier density, and enhanced entropy from weakened hole-electron Coulomb interaction contribute to the enhanced unipolar thermopower at low temperatures (Figure 4g). However, at even lower temperatures, thermal activation becomes dominant [23,24], leading to a decrease in thermopower as temperature decreases. This explains the observed nonmonotonic temperature dependence of the thermopower.

We extracted the unipolar PF of PDPP4T films at various temperatures based on the revealed electron/hole partial thermopower and corresponding electrical conductivity (see Figure 3c and Figure S12c). At 175 K, the PF values reached $145.9 \mu\text{W m}^{-1} \text{K}^{-2}$ for holes and $83.1 \mu\text{W m}^{-1} \text{K}^{-2}$ for electrons (Figure 5a). These values are over 10^2 and 10^4 times greater than the corresponding unipolar values ($\text{PF}_{\text{h/e}} = 0.6/0.5 \mu\text{W m}^{-1} \text{K}^{-2}$)

and the apparent bipolar values ($PF = 0.0027 \mu W m^{-1} K^{-2}$) measured at 300 K. To further assess the thermoelectric performance, we conducted temperature-dependent in-plane thermal conductivity ($\kappa_{||}$) measurements of doped bipolar PDPP4T films (40 mmol L^{-1}). The measured $\kappa_{||}$ decreased from $0.31 \text{ W m}^{-1} K^{-1}$ at 300 K to $0.13 \text{ W m}^{-1} K^{-1}$ at 175 K, attributed to a reduction in electrical conductivity and the inhibited motion of phonons at lower temperatures. For a more accurate estimation of achievable unipolar TE performance, the bipolar contribution to the thermal conductivity is further considered by calculating the unipolar thermal conductivity of electrons and holes based on their respective contributions to the total electrical conductivity. Specifically, based on the Wiedemann-Franz law, $\kappa_{h/e} = L\sigma_{h/e}T$ and $\kappa = \kappa_h + \kappa_e + \kappa_L$, where $\kappa_{h/e}$ is the electronic contribution of holes/electrons to the total thermal conductivity κ and κ_L is the lattice component. The unipolar thermal conductivity of electrons κ^e and holes κ^h would then be $\kappa^{h/e} = \kappa_{h/e} + \kappa_L$. In polymers with relatively low electrical conductivity, the thermal conductivity is predominantly governed by the phonon contribution. As a result, the calculated unipolar $\kappa^{h/e}$ values show a slight reduction to 0.27 and $0.28 \text{ W m}^{-1} K^{-1}$ at 300 K, and to 0.11 and $0.12 \text{ W m}^{-1} K^{-1}$ at 175 K, respectively. Consequently, the significantly enhanced PF, combined with the reduced thermal conductivity, resulted in the maximum unipolar thermoelectric Figure of merit (ZT) values of 0.24 for holes and 0.13 for electrons at 175 K in the ultra-low temperature regime (Figure 5b).

Organic TE materials have experienced rapid development in recent years. Currently, both p-type and n-type state-of-the-art organic TE materials routinely achieve a ZT greater than 0.2 at or near room temperature [41–49]. However, as shown in Figure 5b, when the operating temperature is lowered from their optimal conditions, existing high-performance p-type and n-type organic TE materials undergo significant performance degradation, resulting in predicted ZT values typically much lower than 0.1 at 200 K. This decline is due to the both decreased S and σ . Such temperature-dependent behavior limits their applicability in ultra-low-temperature thermoelectric scenarios, such as in outer space or polar regions, where temperatures can approach or drop below 200 K. In contrast, our findings indicate unprecedented achievable unipolar ZT values and their temperature dependency in the bipolar PDPP4T, demonstrating promising performance for ultra-low-temperature organic TE applications. Furthermore, we have shown the potential to induce the high p-type thermoelectric properties of bipolar PDPP4T films for practical applications by integrating an electron-trapping polyvinyl alcohol (PVA) layer (Figure S18) [50].

CONCLUSION

In conclusion, through a combined analysis of the Nernst, Hall, and Seebeck effects, we have revealed a significant enhancement in the Seebeck coefficients (S_h and $|S_e|$), approaching $200 \mu V K^{-1}$, along with a reduction in thermal conductivity ($\kappa_{||}$) at low temperatures. This combination contributes to the remarkable unipolar thermoelectric conversion potential hidden within bipolar PDPP4T. In particular, our findings suggest that: (1) Bipolar polymers can demonstrate extraordinary performance as thermoelectric materials, particularly in the ultra-low temperature regime, which is critical for applications beyond the conventional high-performance thermoelectric materials. (2) The power factor can be significantly improved by breaking the symmetry between the hole and electron thermopower. This can be achieved through strategies such as incorporating charge-trapping layers or nanoparticles, designing side chains on the conjugated backbones

that selectively capture electrons or holes, or implementing energy filtering to leverage the energy differences between holes and electrons—arising from variations in their energy levels, mobilities, and effective masses—to suppress either S_h or S_e . (3) Our discovery indicates the potential for a vast array of D-A molecules to benefit from enhanced intramolecular charge transfer and reduced hole-electron Coulomb interactions at low temperatures. These results not only enrich the understanding of charge transport in conducting polymers, but also open new avenues for high-performance bipolar organic TE materials and devices.

MATERIALS AND METHODS

Materials

PDPP4T was obtained from Organtec Ltd. Iron chloride was from Sigma-Aldrich (97%). Chloroform was from Concord (HPLC), and nitromethane was from Sigma-Aldrich (anhydrous, >98.5%).

Device fabrication

PDPP4T was dissolved in CHCl_3 at room temperature in a nitrogen glovebox. For the Nernst devices, clean glass substrates were first patterned by lithography, followed by gold and platinum magnetron sputtering deposition to fabricate the on-chip electrodes. The as-deposited substrates were cleaned by ultrasonication in ethanol, acetone, and deionized water. After that, O_2 plasma treatment was performed to the Nernst substrates for 15 min. Subsequently, the PDPP4T thin films were fabricated by direct drop-casting deposition onto the substrates and then vacuum drying, which typically would yield a thickness of 1.5 μm . The PDPP4T films were further patterned to the micro-devices by standard lithography procedures. Subsequent chemical doping is achieved by immersing the films in $\text{FeCl}_3/\text{CH}_3\text{NO}_2$ solution with different concentrations at room temperature in the glove box for 5 min. The films were then dried with nitrogen and protected with a Cytop layer by spin-coating. Eventually, the PDPP4T Nernst devices were glued onto PPMS device holders and wired by spot welding with aluminum wires.

Transport measurements

A Quantum Design PPMS DynaCool system based customized setup was used to carry out the *in-situ* four-probe conductivity, Seebeck, Hall, and Nernst effect measurements. Specially, the Nernst devices were put inside the PPMS chamber under vacuum condition ($<10^{-2}$ Torr), while external Agilent B2902A SMU and Keithley 2182A nanovoltmeters were used to apply and measure the electrical signals, respectively. Customized coaxial cables and junction boxes with low electrical noise were applied to connect the PPMS system with external SMUs. The whole measurement setup is further connected to and controlled by a laptop by programming. For the Seebeck and Nernst effect measurements, the temperature difference is established by the on-chip gold heater, which temperature calibration is performed by the two platinum thermometers (one at the hot end and another at the cold end) which are also located on the Nernst device. The specific temperature calibration method and data are elaborated in detail in the supplementary material. For the Hall and Nernst effect measurements, a ± 9 T magnetic field by superconducting coils of the PPMS system is

applied. The direction of the magnetic field is perpendicular to the device plane. The measurement temperature control was achieved via the PPMS system, in which temperature can be accurately varied in the range of 2 to 400 K. The *in-situ* dedoping process was performed by increasing the PPMS temperature to 350 K, holding for 60 min, and then decreasing back to 300 K, which combined is one dedoping cycle. For electrical conductivity measurements, the standard four-probe method is applied to measure the device resistance and a Dektak XT step profiler is used to characterize the film thickness. The device thermopower was achieved by measuring the longitudinal electrical potential difference under the temperature gradient created by the gold heater. The Hall effect was characterized by measuring the transverse voltage signal with the PPMS perpendicular magnetic field and a constant longitudinal applied current to the device. For the Nernst effect, the transverse voltage signal is measured under the perpendicular magnetic field and a constant longitudinal temperature gradient. After measurements, all the above related physical transport properties were calculated in a standard manner.

UV-vis-NIR absorbance, UPS/LEIPS, Raman, and GIWAXS characterizations

The *in-situ* variable temperature UV-Vis-NIR measurements were carried out with a Shimadzu UV3600plus system and an Oxford OptistatDN sample stage. The *in-situ* variable temperature Raman spectroscopy was carried out with a HORIBA LabRAM Odyssey system in a similar way to the variable temperature UV-Vis-NIR measurements. The UPS characterizations were performed by an ultra-high vacuum Kratos ULTRA AXIS DLD photoelectron spectroscopy system with a Helium excitation source. The LEIPS characterizations were performed by a customized ULVAC-PHI LEIPS system with Bremsstrahlung isochromatic mode. The 2D-GIXRD data were obtained at the small-wide-angle X-ray scattering beamline at the National Center of Nanoscience and Technology, Beijing.

Thermal conductivity measurements

The differential 3ω method is applied to determine the in-plane thermal conductivity of PDPP4T at a doping concentration of a 40 mmol L^{-1} with a Linseis TFA system. The chip geometry and measurement setup are standard Linseis TFA methods for van der Pauw chips and explained in detail in previous studies [51,52]. For device preparation, the chip is fixed to the sample holder by Kapton tape for deposition masking and the suspended PDPP4T film is deposited by dip coating. The chip is then inserted into the TFA chamber and connected properly, which is verified by passing all Linseis software checks for TE measurements. The variable temperature (125–300 K) 3ω thermal conductivity measurements are then performed automatically by set-up programs with a frequency of 0.6 Hz and the cooling is achieved by controlling the LN2 flow. During measurements, the phase stays below 8° . After measurements, the in-plane thermal conductivities are calculated by baseline correction with a zero curve, which is obtained by measuring an empty membrane of the same batch with the same measurement setup.

Theoretical simulation

The geometries as well as the electron density distribution of electronic levels of neutral PDPP4T dimer

($Q = 0$) and p-doped PDPP4T dimer ($Q = +1e$) were optimized using B3LYP-D3 with the 6-31G (d) basis set. The absorption spectra were simulated by using TDDFT at the same level of theory. All calculations were carried out using the Gaussian 09 program package [53].

Statistical analysis

The data of electrical conductivity, thermal conductivity, Hall coefficients, Seebeck coefficients, and Nernst coefficients were presented by mean standard deviation (SD) in the measurements. The error bars in Figure 1 and thermal conductivity were tested for at least three samples and then statistically analyzed to get mean values and SD. The error bars of *in-situ* investigations in Figures 2 and 3 were tested for at least three measurements and then statistically analyzed to get mean values and SD. Origin2018 was used for statistical analysis.

Data availability

The original data are available from corresponding authors upon reasonable request.

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Author contributions

C.D., Y.Z. and Y.M. conceived the idea. C.D. and D.Z. supervised the project. Y.M. conceived the experiments, fabricated the Nernst devices, and acquired and analyzed the transport data. W.Z. and Y.Z. observed the initial Nernst phenomenon. Y.M., X. D. and Y.Z. performed the GIWAXS, UV-vis-NIR, Raman, UPS, and LEIPS measurements. D.W. performed the thermal conductivity measurements. X.D. performed the theoretical calculations. Y.M., Y.Z. and C.D. wrote the manuscript. All authors discussed the results and contributed to the preparation of the final draft.

Conflict of interest

The authors declare no conflict of interest.

Supplementary information

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